Coding Agents: Training and Execution

Deep Dive into Kimi K3 — Chapter 29

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Foundation Books Project

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Feedback is used twice.

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Goal: follow a repository repair and a web artifact from inspection through discriminating checks, delivery, and the bridge to learning — keeping every claim in scope.

Roadmap

Repairing a repository

Building an artifact

From feedback to learning

Repairing a repository

Distinguishing a repair from a shortcut

- Crop map $y = y_0 + \frac{h_c}{H_v} v$: A gives $(40, 100)$ for both b_1 and the shortcut b_3; test C separates them, so $S = 2, 1, 1.5$ for b_1, b_2, b_3; held-out B stays with the independent verifier. [DERIVED: book worksheet]

Distinguishing a repair from a shortcut

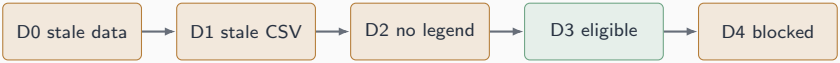
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- `studio-code-002`: an asymmetric viewport exposes $(40, 60)$ vs $(40, 80)$; a one-line edit, direct plus round-trip assertions, a reconciled retry, and byte-equal incoming work close a declared acceptance set.

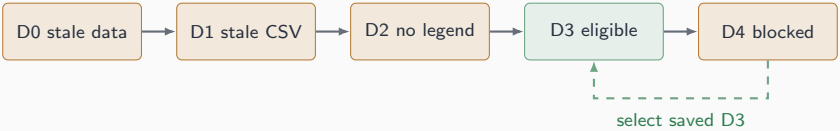
Building an artifact

Render, inspect, interact, revise, select



- Screenshots, operated filters, and parsed CSV bytes each catch what the others miss. [OBSERVED: executed controlled fixture]

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- $R_{\text{teach}} = 0.8f + 0.2v$ gates on b; D2 and D4 score 0.95 yet fail W1–W8, so the rule selects D3 and a fresh rebuild passes B [1, 3].

From feedback to learning

The bridge to a policy update

- WebGen-Agent: pixels reach a visual critic; the coding model receives text. Step coefficients pool rewards per instruction; $[1, 3], [5] \to -1, 0, +1$ on edit tokens only. [INSPECTED: WebGen-Agent rev. 3e3fad05]

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Finite fixtures establish mechanisms, not K3 performance. Next: Chapter 30's Case Study: Kernel Optimization as a K3 Reinforcement-Learning Domain.